

AGENCYOS

Final Gap-Closure Specification

Phase 1 Lead-to-Close Agent Workforce
Implementation Gaps - Dependency Order - Verification - Release Gates

Document	Final Gap-Closure Specification
System	AgencyOS
Scope	Phase 1 Lead-to-Close
Locked agents	Sales Scheduler Quotation Master Coordination Orchestrator/Router
Purpose	Close architectural and implementation gaps before production
Prepared	2026-09-11

Core rule: Code existing is not the same as feature complete. A gap is closed only when the capability works through the approved contract, passes required tests, preserves evidence, and satisfies its Definition of Done.

1. Executive Objective

This document is the final control layer for converting the already-defined AgencyOS Phase 1 architecture into a verifiable implementation. The target is a complete Lead-to-Close journey: lead intake, identity/deduplication, Sales qualification, scheduling when required, structured requirements, policy-compliant quotation, governed approval, controlled negotiation, exact-version acceptance, payment/exception gate and reliable WON state.

AgencyOS requires evidence-based completion, explicit state transitions, policy-controlled authority, exact version traceability, safe retries, auditable approvals and truthful failure states. The technical contract layer adds typed operations, stable identities, schema validation, idempotency, tenant isolation and normalized provider integrations.

2. Locked Phase 1 Boundary

Agent	Owns	Must not own
Sales	Lead, qualification, discovery, trust, objections, negotiation within policy, conversion evidence	Provider routing, policy override, payment verification, unsupported commitments
Scheduler	Intent, availability, slot proposal, booking, reminders, reschedule/cancel, completion evidence	Invent availability, approve commercial terms, verify payment
Quotation Master	Quote construction, pricing rules, tax configuration, milestones, versioning, approval/delivery workflow	Silent edits to accepted baseline, invented pricing/tax, policy override
Coordination	Workflow state, dependencies, handoffs, blockers, next workflow step	Model/provider selection or domain-commercial decisions
Orchestrator/Router	Capability routing, model/provider selection, execution control, retry/fallback, result validation	Business approval, pricing authority, requirement invention
Policy/Admin	Authority, exceptions, approval decisions, governance	Routine agent execution

The architecture explicitly separates routing/execution control from specialist domain responsibility and from business approval authority.

3. Master Gap-Closure Matrix

Use this as the implementation control board. Each gap remains OPEN until code, configuration, test evidence and state/audit evidence are present.

ID	Gap	Owner	Dependency	Closure evidence
G-01	Repository baseline and evidence audit	Implementation	None	Audit + build/test baseline
G-02	Tenant, auth and RBAC enforcement	Platform	G-01	Authorization + isolation tests
G-03	IDs, audit and correlation primitives	Platform	G-01	Schema + trace evidence
G-04	Task/result/handoff/event contracts	Orchestrator/Coordination	G-03	Contract tests + traces
G-05	Domain state machines	Backend	G-03	Transition + DB tests
G-06	Policy/approval enforcement	Policy/Admin	G-02,G-05	Approval + enforcement tests
G-07	Idempotency/retry/timeout	Workflow infra	G-03,G-04	Failure/retry tests
G-08	WhatsApp adapters and dedupe	Integration	G-02,G-07	Webhook evidence
G-09	Sales/requirements flow	Sales	G-04,G-05,G-08	Sales E2E trace
G-10	Scheduler booking lifecycle	Scheduler	G-04,G-07	Booking + provider ID
G-11	Quote/version/approval lifecycle	Quotation	G-05,G-06,G-09	Exact-version evidence
G-12	Workflow/dependency tracking	Coordination	G-04,G-05	Task graph + handoffs
G-13	Capability/model/provider routing	Orchestrator	G-04,G-06	Routing + fallback trace
G-14	Provider adapters	Integration	G-07,G-13	Adapter contract tests
G-15	Observability/audit	Platform	G-03,G-07	Queryable trace
G-16	360 QA/security/regression	QA	All functional	Test report + defects
G-17	Release/smoke/rollback	DevOps/QA/Admin	G-16	Exact build + approval

4. Dependency-Ordered Implementation

Stage	Build sequence	Exit condition
0 - Baseline	Repository, stack, routes, DB, tests, integrations, config, TODO/mocks/broken inventory	Exact current state documented
1 - Foundation	Tenant, auth/RBAC, IDs, audit, entities, state machines	Core state is enforceable
2 - Contracts	Task, result, handoff, event, approval, quote/version, provider schemas	Contract tests pass
3 - Workflow infra	Queue/jobs, idempotency, retry, timeout, DLQ, correlation	Durable recovery works
4 - Agents	Sales, Scheduler, Quotation, Coordination, Orchestrator	Agent DoD passes
5 - Channel/commercial	WhatsApp, CRM mapping, approvals, quote delivery, acceptance, payment gate	Lead-to-Close works
6 - Verification	Unit, API, DB, integration, E2E, security, regression, performance	Required gates pass
7 - Release	RC, smoke, monitoring, rollback, Admin approval	Exact tested build approved

The roadmap requires repository inspection before rewriting, evidence-based gap analysis, dependency-ordered implementation and verification at every phase.

5. Contract Closure

Every important AgencyOS action must be a typed, authorized, traceable operation with stable identity, explicit scope, validated input, structured output/evidence and safe failure behavior.

Universal contract checklist

- request_id and correlation_id are present; side-effecting operations use idempotency_key.
- organization_id and resource scope are explicit for tenant work.
- task_id, event_id, approval_id, quote_id/version and other identities are stable.
- Task inputs and agent/provider outputs are schema-validated before acceptance.
- Provider responses are untrusted until normalized and validated.
- External IDs map to stable internal IDs; external state is separate from domain state.
- Authorization and policy checks occur before high-risk execution.
- Canonical errors replace false-success or empty-state behavior.
- Critical operations retain evidence and audit trace.

Required task lifecycle

CREATED -> VALIDATING -> READY -> DISPATCHED -> ACKNOWLEDGED -> IN_PROGRESS -> WAITING_FOR_RESULT -> RESULT_RECEIVED -> VALIDATING_RESULT -> ACCEPTED -> HANDOFF_READY -> HANDED_OFF -> VERIFIED -> CLOSED

Exception states: BLOCKED, RETRYING, FAILED, EXPIRED, CANCELLED, ESCALATED. Result receipt does not equal acceptance.

6. Agent-by-Agent Closure

Sales Agent

- Lead creation and deduplication map to one authoritative CRM lifecycle.
- Qualification and requirements are structured; assumptions remain separate from confirmed facts.
- Trust/objection playbooks use approved evidence and commercial boundaries.
- Pricing, discounts, payment terms and commitments remain policy-controlled.
- Acceptance references the exact quote version and is never inferred from ambiguous text.
- WON requires configured evidence; LOST requires a valid reason.

Scheduler Agent

- Normalize intent, mode, date, time and timezone before availability lookup.
- Availability comes from the authoritative calendar/resource source.
- Proposal and booking are distinct states; booking re-checks availability.
- Provider event ID/reference is persisted and booking is idempotent.
- Reschedule/cancel preserves history and replaces obsolete reminders.
- Completion is explicit and authorized; time passing alone is not completion.

Quotation Master Agent

- Uses confirmed requirement version and active commercial policy.
- Calculates price, discount, tax and totals deterministically from stored inputs.
- Material commercial changes create immutable new quote versions.
- Approval, delivery and acceptance reference the exact version.
- Out-of-policy actions stop and request approval.
- Accepted historical versions are never silently edited.

Coordination Agent

- Durable workflow instances, current step, dependencies, owners and blockers.
- Completed tasks require accepted result/evidence.
- Blocked tasks cannot silently advance.
- Tracks workflow handoffs and escalation.
- Does not select models/providers or make commercial decisions.

Orchestrator/Router Agent

- Resolves capability and eligible specialist agent.
- Checks permissions, data classification and tool requirements.
- Selects model/provider using configured routing, health, capability, cost and latency inputs.
- Records route, model/provider and policy version.
- Fallback preserves the same organization/data restrictions.
- Validates result schema and bounds retries.
- Cannot approve commercial exceptions or rewrite policy.

7. Failure and False-Success Closure

Failure handling is a release-critical capability. AgencyOS must never convert a timeout, outage, duplicate webhook, malformed AI response, ambiguous acceptance or payment mismatch into a successful business state.

Failure	Required behavior	Evidence
Provider timeout	Safe retry or fallback; verify state	Retry trace; no duplicate side effect
Provider outage	Approved fallback with same policy/data restrictions	Fallback record
Rate limit	Backoff/alternate; no storm	Retry history
Invalid credentials	Stop and alert	Incident record
Schema failure	Validate; correct/alternate; reject invalid	Validation evidence
Duplicate webhook	Deduplicate by provider event key	One business effect
Concurrent quote edit	Conflict/version control	Version/conflict evidence
Ambiguous acceptance	Clarify; hold	No WON transition
Policy ambiguity	Admin escalation	Approval/escalation record
Payment mismatch	Hold and verify	No paid transition
Repeated agent failure	Bound retry; reassign/escalate	Failure summary
WhatsApp unavailable	Queue/alternate by policy	Pending/error trace

8. 360-Degree Verification

Phase 1 requires functional, UI where applicable, API, database, integration, E2E, regression, security, performance, compatibility and deployment/smoke verification. Hard gates override aggregate readiness scores.

Test	Scenario	Expected evidence
F-01	New WhatsApp lead maps to correct lead	Lead ID + audit
F-02	Existing identity deduplicates	Match evidence
F-03	Qualification becomes structured	Qualification record
F-04	Requirements become versioned	Requirement version
F-05	Meeting intent routes to Scheduler	Task/result trace
F-06	Retry creates one booking	Idempotency evidence
F-07	Requirements create quote draft	Quote version
F-08	Out-of-policy quote requests approval	Approval ID
F-09	Approved quote delivered exactly	Version evidence
F-10	Negotiation creates immutable version	Version history
F-11	Ambiguous acceptance does not close	No WON
F-12	Exact acceptance references exact quote	Acceptance evidence
F-13	Payment/exception controls WON	Verification/exception
F-14	WON preserves sales context	Handoff evidence
F-15	Provider failure is controlled	Error trace

Mandatory negative tests

- Malformed task/result payloads; unauthorized agent/tool calls; cross-tenant access.
- Duplicate webhook events; expired approvals; old quote delivery; concurrent revisions.
- Calendar race conditions; provider timeout after possible side effect.
- AI output attempting to grant permission; ambiguous acceptance; unverified payment proof.

9. Release Candidate Gates

Gate	Must pass	Blocker
Build	Compiles/packages; artifact identified	Build failure
Contract	Schemas/API contracts	Breaking contract
Security	Auth/RBAC/tenant/secrets/injection	Critical finding
Functional	Critical Phase 1 scenarios	Critical path failure
Integration	WhatsApp/calendar/AI adapters	False success/data corruption
Regression	Supported behavior preserved	Critical regression
Performance	Configured critical thresholds	Unacceptable degradation
Observability	Trace/audit/error visibility	Untraceable critical operation
Rollback	Applicable recovery path validated	Unsafe recovery
Approval	Required Admin decisions exist	Missing approval
Smoke	Post-deploy critical path succeeds	Smoke failure

Skipped critical tests are not passes. Compilation is not functional correctness. The approved release candidate must be the exact build that was tested and deployed.

10. Final Phase 1 Acceptance Checklist

- Repository baseline and gap analysis are documented.
- Tenant isolation and permissions are backend-enforced.
- IDs, task/correlation IDs and audit events are persistent.
- Task/result/handoff/event schemas are implemented and tested.
- State transitions are explicit, valid and auditable.
- Required side effects are idempotent; retries are bounded and operation-aware.
- Provider integrations use adapters and normalized responses.
- WhatsApp webhooks authenticate and deduplicate.
- Sales lifecycle and requirements are structured and traceable.
- Scheduler booking is timezone-aware and idempotent.
- Quotation pricing/discount/tax/payment behavior is policy-controlled.
- Material quote changes create immutable versions.
- Approval and acceptance reference exact versions/actions.
- WON is blocked until configured payment/exception evidence exists.
- Coordination tracks dependencies, blockers and accepted handoffs.
- Orchestrator records routing/model/provider decisions and safe fallback.
- Provider failure cannot produce false success.
- 360-degree tests and critical E2E scenarios pass.
- Security and tenant isolation tests pass.
- Release candidate maps to exact source/build identity.
- Deployment smoke tests pass.
- Audit timeline is queryable.
- No unresolved critical blocker or API/data-contract ambiguity remains.

11. Required Evidence Package

Category	Minimum evidence
Repository	Commit/version, branch/tag, dependency state
Build	Build ID, artifact reference/checksum, environment reference
Tests	Unit, contract, API, DB, integration, E2E, regression, security, smoke
Workflow	Representative task/event/handoff traces with correlation IDs
Routing	Capability match, agent, model/provider, routing policy/version
Commercial	Quote versions, policy decisions, approval IDs, acceptance evidence
Scheduler	Availability, proposal, booking/provider reference, reminder state
Failure	Retry/fallback/escalation traces
Security	Auth/RBAC/tenant/secrets findings
Release	Readiness gates, Admin approval, deployment and smoke results

The evidence package must allow an Admin or reviewer to reconstruct what happened, which agent/model/provider acted, which exact version was involved, what policy applied, what evidence proves it, and what happened on failure.

12. Final Gap-Closure Rulebook

Rule	Implementation meaning
AI is not authority	AI executes within policy; backend controls authority
WhatsApp is not source of truth	Messages are evidence; structured domain state is authoritative
Provider is replaceable	Provider-specific behavior stays in adapters
Version is immutable	Material commercial change creates a new version
Approval is specific	Approval authorizes an exact action under exact conditions
Acceptance is exact	Client acceptance identifies exact commercial version/terms
State is truthful	Failure remains pending, blocked or escalated
Retry is safe	Never blindly replay non-repeatable/high-risk side effects
Evidence is mandatory	Completion requires objective evidence where configured
Scope is protected	Material new work follows revised scope/change workflow
Admin remains authority	Commercial exceptions and high-risk actions remain governed

FINAL PHASE 1 DEFINITION OF DONE: A complete Lead-to-Close journey executes repeatedly on the approved release candidate with structured agent collaboration, exact-version commercial control, reliable scheduling, policy-gated exceptions, durable workflow state, authenticated/idempotent integrations, objective QA evidence and no known critical gap requiring manual database intervention for ordinary progression.

Reference basis: AgencyOS Master Vision, Complete Business Workflow, AI Agent Architecture, Orchestrator/Routing, CRM/Sales, WhatsApp, API/Integration, Automation/Jobs, Development, QA/Production Readiness, Admin Policy, Onboarding and the Phase 1 agent responsibility/interaction specifications.